DATA MINING PROJECT

Master in Data Science and Advanced Analytics

NOVA Information Management School

Universidade Nova de Lisboa

Clustering Report

Amazing International Airlines Inc.

Group 92

Duarte Gomes, 20250017

Esra Salhi, 20250537

Mehmet Karaca, 20250344

Fall/Spring Semester 2025-2026

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1. EXECUTIVE SUMMARY

This analysis delivers a portfolio-based customer segmentation framework that identifies where Amazing International Airlines Inc. should focus investment to maximize total customer value. The final output consists of nine actionable customer segments derived from the integration of value-based and demographic perspectives, enabling differentiated customer treatment based on both economic potential and scale.

The segmentation reveals that long-term profitability will not be driven by a small group of very high-CLV customers, but by large, high-income segments with moderate and currently underutilized lifetime value. The most significant growth opportunity lies in urban, high-income customer groups, including approximately 2,500 customers in the High-Income | Urban, Single/Divorced segment. Despite strong purchasing power, these customers currently exhibit CLV levels close to the population average, indicating substantial unrealized value. Due to their size, even small improvements in engagement and loyalty would generate a disproportionate impact on total portfolio value.

Based on these findings, the recommended strategy is to shift focus from niche optimization toward scalable value creation. Large, under-monetized high-income segments should be prioritized through targeted commercial actions and loyalty mechanisms that emphasize convenience, flexibility, and perceived value rather than price competition. Differentiated fare bundles, flexible ticket conditions, premium service access, and loyalty incentives designed to increase travel frequency are well aligned with the needs of these segments.

High-CLV customers, while strategically important, should be managed with a defensive focus on retention and risk mitigation. Consistent service quality, recognition-based communication, and stable premium benefits are sufficient to preserve their contribution without positioning them as the primary growth engine. Low-value segments, in contrast, should be managed with strict cost discipline through standardized services and automated communication to avoid inefficient allocation of resources.

Overall, the proposed segmentation framework supports a decisive shift from uniform customer treatment to a portfolio-driven strategy. By prioritizing customer segments according to both economic potential and scale, AIAI can concentrate investment where returns are highest, protect existing profitability, and manage low-impact segments efficiently, creating a scalable and economically grounded foundation for long-term growth.

2. INTRODUCTION

Building on the exploratory analysis conducted in Phase 1, this report advances from data understanding to clustering-based customer segmentation in order to support more informed and actionable strategic decisions for Amazing International Airlines Inc. (AIAI). The exploratory phase revealed several structural and behavioral patterns that directly shaped the objectives and design choices of the clustering phase.

In particular, Phase 1 highlighted a weak relationship between income and Customer Lifetime Value (CLV), a strong geographic concentration of customers in Ontario, British Columbia, and Quebec, and pronounced heterogeneity in travel activity and engagement levels. These findings indicated that customer value cannot be adequately explained by single variables and motivated the need for a multi-dimensional segmentation approach capable of capturing both economic relevance and scale.

Based on the data quality issues identified during Phase 1, a dedicated data preparation and feature engineering pipeline was implemented prior to clustering. This included correcting logical inconsistencies in the transactional data, handling missing values for income and CLV, and transforming raw variables into analytically meaningful features. All preprocessing and feature engineering steps applied in the clustering phase represent a direct operationalization of Phase 1 insights and are fully documented in the preprocessing notebook.

Feature engineering focused on variables with demonstrated relevance during exploratory analysis, including indicators of customer tenure, activity status, and engagement with the loyalty program. Continuous variables were standardized and categorical attributes were encoded to ensure consistency and robustness in distance-based clustering methods.

The objective of the clustering phase was therefore not to maximize technical complexity, but to derive interpretable and actionable customer segments that support differentiated marketing, loyalty design, and resource allocation decisions. By translating exploratory insights into a structured segmentation framework, this phase provides the analytical foundation for a portfolio-based customer strategy.

3. METHODOLOGY

Feature Selection Process

The feature selection process was grounded in insights obtained during Deliverable 1 (Exploratory Data Analysis), where the structure, quality, and relevance of available variables were systematically assessed. This phase provided an initial understanding of which features carried meaningful information for segmentation and which introduced noise, redundancy, or interpretability challenges.

At the beginning of the clustering phase, a trial-and-error approach was adopted. Multiple feature combinations and segmentation perspectives were tested to explore data structure and evaluate the trade-off between model performance, interpretability, and business relevance. In line with the project guidelines, three segmentation perspectives were initially defined: behavioral, value-based, and demographic.

The value-based perspective aimed to capture customers' economic contribution and profitability. Core features included Customer Lifetime Value and Income, complemented by loyalty program indicators such as total points accumulated and total monetary value of redeemed points, reflecting both long-term value and engagement with the airline's loyalty ecosystem.

The demographic perspective focused on customer characteristics such as Gender, Location Code, Education, and Marital Status. As all demographic variables were categorical, one-hot encoding was applied to make them suitable for distance-based clustering algorithms.

The behavioral perspective focused on customer flight activity and travel patterns using the transactional flight database. Due to its monthly structure, the data was aggregated at the customer level to avoid duplicate records and to derive behavioral features such as total number of flights, total distance flown, and companion travel intensity.

Multiple clustering algorithms, including k-means++ and hierarchical clustering, were applied and evaluated using standard internal validation metrics. While technically valid solutions were obtained, the resulting clusters showed limited stability and weak business interpretability.

Since the behavioral perspective did not provide additional, actionable differentiation it was excluded from the final segmentation framework. This decision ensured a more interpretable and business-relevant solution without introducing redundant segmentation dimensions.

During initial experimentation, a large feature set was used to maximize clustering metrics. However, both empirical results and business considerations showed that higher dimensionality negatively affected cluster stability and interpretability, highlighting the curse of dimensionality. As a result, the feature space was refined, and the final approach focused on two complementary perspectives: demographic and value-based segmentation, which provided the most stable, interpretable, and business-relevant results. These perspectives were later combined into the final integrated segmentation framework. Final feature sets:

Demographic Features: Location Code (Urban, Suburban), Gender, Education Level, Marital Status

Value-Based Features: Customer Lifetime Value, Income, Total Points Accumulated, Total Cost of Points Redeemed

This reduced, perspective-driven feature selection ensured a balance between analytical robustness, interpretability, and strategic relevance.

Algorithm Selection

Algorithm selection followed an iterative, evaluation-driven approach. Initially, multiple clustering algorithms were applied across all three perspectives. While several methods achieved strong metric performance, many resulted in clusters with limited business interpretability.

Consequently, the focus shifted from metric optimization to interpretability, stability, and business relevance. The final comparison was restricted to k-means++ and hierarchical clustering using different linkage criteria (single, average, complete, and Ward). These methods demonstrated a favorable balance between clustering quality, computational efficiency, and interpretability. Hierarchical clustering additionally enabled dendrogram-based visualization to support the selection of an appropriate number of clusters.

Both methods were applied to the demographic and value-based perspectives. From a business standpoint, the number of clusters was restricted to a range of two to ten, as higher values would reduce usability in a managerial context. For each method, all feasible values of k were evaluated using multiple technical metrics, including R^2 score, Silhouette score, Davies–Bouldin index, and Inertia, supported by visualization.

A key decision threshold was defined for the R^2 score, with values above 0.5 considered sufficient for business objectives. Among these, preference was given to solutions with fewer clusters to enhance interpretability and strategic applicability. The Elbow Method was applied to identify points of diminishing returns, while the remaining metrics were used primarily as validation tools to ensure cluster stability and structural soundness.

Parameter Tuning

For k-means clustering, the k-means++ initialization was used to ensure well-separated initial centroids and improve convergence stability. To reduce sensitivity to random initialization, each configuration was executed with n_init = 20, and a fixed random_state = 42 was applied to guarantee reproducibility of results.

Euclidean distance was used consistently across all distance-based clustering methods. Prior to modeling, all continuous variables were standardized using z-score normalization to ensure comparable feature scales, while categorical variables were transformed using one-hot encoding. This preprocessing pipeline ensured consistent distance computation and prevented scale-driven bias in cluster formation.

Implementation Approach and Comparison of Clustering Methods

A key challenge in the implementation phase was integrating multiple segmentation perspectives into a single coherent framework. To address this, a cross-tabulation approach was adopted, combining cluster assignments from the demographic and value-based segmentations. This approach preserves the interpretability of each perspective while enabling joint strategic analysis.

The demographic clustering resulted in six clusters, while the value-based clustering produced four clusters, leading to a theoretical maximum of twenty segment combinations. Several alternative integration strategies were evaluated, including feature concatenation, weighted distance approaches, and multi-view clustering. However, these methods either increased dimensionality at the expense of interpretability or generated segments that were difficult to translate into actionable business profiles.

To ensure managerial usability, segment combinations with very small sample sizes or limited strategic relevance were excluded. For example, customers characterized by both low income and low customer lifetime value were deprioritized due to their limited revenue potential. After this refinement, the final segmentation consisted of nine distinct and well-differentiated customer segments.

Overall, this implementation approach prioritizes interpretability and strategic usability over methodological complexity, resulting in a segmentation framework that balances analytical rigor with practical business relevance.

4. RESULTS & VALIDATION

Technical metrics and statistical significance – Value-Based Perspective

The evaluation of clustering performance for the value-based perspective was based on a combination of internal validation metrics, including R^2 score, Silhouette score, Davies–Bouldin index, and Inertia (Figure 1, Annex).

These metrics were jointly considered to balance explanatory power (R^2) with cluster cohesion and separation, ensuring that the selected solution is both statistically sound and business-interpretable.

The R^2 score increases with the number of clusters, showing a clear convergence pattern. A four-cluster solution was selected, achieving an R^2 score of 0.5215, indicating that approximately 52% of the variance in value-related customer behavior is explained. From a business perspective, this level of explained variance was considered sufficient, as customer behavior is inherently heterogeneous.

The selected solution achieved a Silhouette score of 0.288, indicating acceptable cluster separation and cohesion for a real-world customer segmentation problem. This result is supported by a Davies–Bouldin index of 1.1763, suggesting a balanced trade-off between within-cluster compactness and between-cluster separation.

Inertia was analyzed primarily through the Elbow Method to identify diminishing returns from increasing the number of clusters. The selected four-cluster configuration aligns with the observed stabilization pattern, supporting its selection as a parsimonious and interpretable solution.

Overall, the combined evaluation of these metrics supports the choice of a four-cluster solution that balances explanatory power, structural quality, and business interpretability. Cluster assignments remained stable across multiple model initializations, indicating that the selected solution is robust and not driven by random starting conditions.

Technical metrics and statistical significance – Demographic Perspective

The demographic perspective was evaluated using the same validation framework to ensure methodological consistency. Based on the observed metric behavior, a six-cluster solution was selected, achieving an R^2 score of 0.60 and indicating a substantial level of explained variance in demographic characteristics (Figure 3, Annex). Using the same validation framework ensured comparability across perspectives and supported consistent decision-making in the integrated segmentation.

The corresponding Silhouette score of 0.3353 and Davies–Bouldin index of 1.0447 indicate reasonable cluster separation and internal consistency. The Inertia curve further supports this choice, showing a clear elbow pattern that reflects an appropriate balance between model complexity and compactness.

Overall, the demographic clustering demonstrates strong technical performance and provides a reliable basis for subsequent business interpretation and strategic analysis.

Customers Segments and Profiles

Based on the applied clustering approaches, distinct customer segments were identified from both the value-based and demographic perspectives. This section interprets the business meaning of the discovered segments and explains the rationale behind the final selection used for strategic analysis.

From the value-based perspective, four initial customer profiles were identified: High Income, Low Points and Costs, High Customer Lifetime Value (CLV), and Low Income. Based on heatmap analysis and domain knowledge, the Low Income segment was excluded from further strategic consideration. Although this group represents a relatively large portion of the customer base, its limited spending capacity implies a low potential for CLV growth. Given the project's objective of maximizing revenue and long-term customer value, this segment was therefore deprioritized.[1]

The High CLV segment, despite its relatively small size, was retained due to its strategic relevance. This group represents the only segment with consistently high lifetime value, making it essential for understanding the key drivers of long-term profitability and for designing targeted retention and loyalty strategies.

For the demographic perspective, six clusters were initially identified: Master Students, Married & Suburban, Single & College, Doctors, Married & Urban, and High School or Below. The distribution of customers across these clusters was uneven, with several segments exhibiting relatively small sample sizes, as illustrated in Figure 4 in the annex. In line with the strategic focus on scalable and actionable customer groups, clusters with very small sample sizes were excluded from further analysis. This decision was guided by domain knowledge and the practical requirement that effective marketing and service strategies should target customer segments with sufficient population size.[2]

After this refinement process, the integration of the remaining value-based and demographic clusters resulted in nine final customer segments. These segments were visualized using a heatmap[3] and bar chart[4], enabling a clear comparison of segment size and characteristics.

The largest customer segment consists of high-income customers who are married and live in suburban areas. This group shows higher engagement with the loyalty program, as reflected by increased point accumulation and redemption activity. Additionally, most customers in this segment hold education levels other than a college degree, suggesting a distinct socio-demographic profile.[5]

The second-largest segment comprises customers with low levels of accumulated and redeemed points who are predominantly married and reside in suburban areas. While their engagement with the loyalty program is limited, their income level is slightly above the overall customer average, indicating untapped potential for increased program participation.[6]

The integration of value-based and demographic clustering has resulted in nine distinct customer segments. These segments provide a comprehensive map of the AIAI customer base, ranging from high-value urban professionals to price-sensitive rural travelers. Figure 6 summarizes the key characteristics of these nine segments, including their average CLV, preferred Loyalty Status (Star/Nova/Aurora), and geographic concentration.[7] This landscape serves as the factual foundation for the strategic interventions detailed in the following chapter.

[1] Figure 2
[2] Figure 4
[3] Figure 5 and 6
[4] Figure 8
[5] Figure 8
[6] Figure 8
[7] Figure 6

5. STRATEGIC RECOMMENDATIONS

To translate the nine segments into an actionable business strategy, we have grouped them into four strategic pillars based on their Customer Lifetime Value (CLV) and Demographic profile. This allows AIAI to allocate resources efficiently—investing heavily where potential is high and maintaining cost-effective operations where value is lower. The following recommendations focus on three mandatory areas: **Marketing Approaches, Service Customization, and Loyalty Program Optimization.**

Pillar 1: Growth Strategy (Focus: High Income / Urban / Single)

- Marketing Approach: Target this segment with "Premium Urban Lifestyle" campaigns. Since they are Single and Urban, marketing should focus on flexibility and spontaneous travel rather than family-oriented deals.
- Service Customization: Offer "Elite-Tier" perks (e.g., lounge access or priority boarding) regardless of their current Loyalty Status for a trial period. High-income individuals are more likely to upgrade if they experience premium service first-hand.
- Loyalty Optimization: Implement a "Fast-Track" to Star Status based on Income level and initial engagement. This capitalizes on their high CLV potential before they choose a competitor.

Pillar 2: Relationship Management (Focus: High Income / Suburban / Divorced or Married)

- Marketing Approach: This is your largest high-value group. Use Location data to offer suburban-specific incentives, such as parking vouchers or "Family & Friends" referral bonuses that appeal to their Marital Status.
- Service Customization: Focus on "Seamless Comfort." Use their high CLV to justify dedicated customer support lines or personalized "welcome back" recognitions during the service process.
- Loyalty Optimization: Prioritize Retention rewards. Since they already provide high value, the goal is to prevent churn by offering milestone rewards (e.g., anniversary bonus points) to maintain their Loyalty Status.

Pillar 3: Efficiency & Retention (Focus: Middle-to-Low Income / Rural & Suburban)

- Marketing Approach: Focus on "Value-for-Money" messaging. Use automated email marketing to minimize costs while highlighting affordable routes.
- Service Customization: Standardize service delivery to maintain low operational costs. Personalization should be limited to basic digital self-service tools.
- Loyalty Optimization: Use a "Points-Only" engagement model. Encourage these customers to redeem points for low-cost upgrades or partner rewards to clear loyalty liabilities from AIAI's balance sheet.

5.1. Recommendation to the CEO

The following strategic summary outlines the shift from uniform customer treatment to a **portfolio-driven growth strategy**. Our analysis confirms that AIAI's long-term profitability will not be driven by optimizing high-value niches, but by unlocking the latent potential of large, under-monetized customer segments.

Core Strategic Shift: Scalability Over Niche Optimization

Currently, AIAI possesses several high-income segments—particularly in Urban and Suburban locations—where the average Customer Lifetime Value (CLV) remains close to the population mean despite significant purchasing power.

The primary recommendation is to prioritize these "Mass-Affluent" segments. Because of their scale (often exceeding 2,000 customers per cluster), even a marginal 5% increase in their individual engagement or loyalty participation will generate a far greater impact on total revenue than doubling the value of a small high-CLV niche

Strategic Implementation Pillars

To operationalize this, we recommend managing the customer portfolio through four distinct pillars:

1. The Growth Engine (High Income | Urban | Single/Divorced): This is our primary target for value expansion. Strategies should focus on perceived value and flexibility—using differentiated fare bundles and premium service trials to convert high income into sustained CLV.
2. Relationship Stabilization (High Income | Suburban | Married): As the largest and most stable segment, the focus here is on retention and household value. We recommend introducing family-oriented loyalty mechanisms, such as "Pooled Points" or joint booking benefits, to secure their long-term loyalty.
3. **Asset Protection (High-CLV Segments):** These customers are strategically vital but have limited growth scale. The objective is **risk mitigation** through consistent service quality and recognition-based communication to ensure they remain within the AIAI ecosystem without requiring expensive acquisition costs.
4. Operational Efficiency (Low-Value Clusters): Segments with low income and low CLV potential should be managed with strict cost discipline. By using automated communication and standardized service levels, AIAI can ensure that commercial resources are not diverted from high-impact growth segments.

By adopting this differentiated approach, AIAI can concentrate its marketing and service investments where the intersection of scale and economic potential is highest. This strategy provides a clear path to scalable growth while protecting existing high-value assets and maintaining operational simplicity.

6. FINANCIAL IMPACT MODELING

This analysis extends the customer segmentation framework by translating segment-level insights into a structured financial impact model. The objective is not to predict exact future revenues, but to evaluate whether targeted customer initiatives are economically justified under realistic operational assumptions.

Value Baseline Construction

A value baseline was constructed at the customer level using a weighted index of three observable drivers: flight frequency (40%), total distance flown (30%), and loyalty program engagement (30%). Flight frequency was intentionally assigned the highest weight, as it most directly captures recurring revenue behavior, while distance and points accumulation reflect usage intensity and engagement rather than pure transactional value.

As the resulting index represents a relative score rather than a monetary measure, it was calibrated into a monetary proxy to ensure coherence with absolute cost figures. After calibration, the average customer baseline value is approximately 300 units per month, with segment averages ranging between 280 and 360 units, providing a realistic economic scale for impact modeling.

Segment-Level Value Distribution

Baseline aggregation reveals that portfolio value is concentrated in a small number of large segments. In particular, High Income | Single, College and Low Income | Urban, Divorced each represent more than 700k units of baseline monthly value, driven primarily by customer volume rather than individual value. This highlights an important structural insight: portfolio growth is more strongly influenced by scalable segments than by niche high-value profiles.

Impact Assumptions and Cost Structure

For each priority segment, explicit assumptions were defined regarding expected uplift, adoption, and costs. Uplift rates range from 2% to 7%, consistent with observed effects in CRM, personalization, and loyalty initiatives, while adoption rates vary between 30% and 55% depending on segment profile. Higher adoption was assumed for high-income and urban segments, reflecting greater responsiveness to targeted value propositions.

Costs were modeled using a two-part structure. Fixed costs represent setup, integration, and campaign design, ranging from 4,000 to 15,000 units, while variable costs capture ongoing per-customer incentives and operational expenses. This structure ensures that both scale effects and operational efficiency are explicitly reflected in the financial outcomes.

To improve operational realism, fixed costs were phased over the initial implementation period and uplift and adoption were allowed to ramp up gradually, reflecting learning effects and delayed behavioral response.

Financial Results and Portfolio Performance

Under conservative assumptions (12-month horizon), the initiative already delivers a positive portfolio NPV of approximately 0.30M units, with payback achieved within two months. Extending the horizon to 24 months increases portfolio NPV to approximately 0.79M units, while the optimized 36-month scenario reaches an NPV of approximately 1.39M units, with payback occurring in the first month.

Value creation is highly concentrated. The High Income | Single, College segment alone generates an NPV exceeding 0.5M units, followed by High Income | Urban, Divorced with approximately 0.17M units. In contrast, smaller niche segments such as High CLV | Doctor fail to recover their fixed costs and generate negative standalone NPVs, despite high individual value. This confirms that customer scale, rather than peak individual value, is the dominant driver of economic impact in this portfolio.

Strategic Implications

The results suggest that the optimal strategy is not to maximize value per customer, but to selectively activate large, under-monetized segments with strong adoption potential. A phased rollout strategy is recommended, prioritizing high-income and urban segments while excluding structurally unprofitable niches. This approach maximizes portfolio-level returns while limiting downside risk and unnecessary capital allocation.

Overall, the financial impact model demonstrates that targeted customer initiatives can generate substantial and rapid value when aligned with segment scale, realistic behavioral assumptions, and disciplined investment selection.

7. CONCLUSION

This report has established a comprehensive, data-driven segmentation framework for Amazing International Airlines Inc. (AIAI). By transitioning from a fragmented view of the customer base to a structured, portfolio-based approach, we have provided the strategic clarity necessary to maximize customer value across nine distinct segments.

Synthesis of Findings and Strategy

The core of our technical findings lies in the successful integration of value-based and demographic clustering. This approach revealed that AIAI’s primary opportunity for growth does not reside in niche high-value clusters, but in large, high-income "Mass-Affluent" segments located in urban and suburban areas. To bridge the gap between technical complexity and operational execution, these segments were consolidated into four strategic pillars: Growth, Relationship, Retention, and Efficiency. This framework allows AIAI to move away from a "one-size-fits-all" marketing model and toward a targeted allocation of resources where investment is proportional to potential return.

Methodological Reflection and Limitations

The chosen methodology prioritized actionability over purely statistical optimization. While several behavioral dimensions were explored, they were intentionally secondary to the demographic-value integration to ensure that the resulting segments were easily identifiable for marketing teams. A limitation of this approach is the static nature of the current segmentation; it captures a "snapshot" of the customer base.

Furthermore, from a statistical standpoint, it must be acknowledged that the models achieved a non-perfect R2 score. This indicates that the clustering results do not capture 100% of the variance within the dataset, and some degree of overlap between segments remains. While this does not compromise the strategic utility of the pillars, it confirms that no model can perfectly mirror the complexities of human travel behavior. Additionally, because behavioral flight patterns were not the primary driver of the final nine segments, some nuances in seasonal travel frequency may remain untapped in the current strategic pillars.

Future Research Directions

To maintain a competitive edge, AIAI should evolve this framework in two specific directions:

1. Predictive Analytics: Future research should focus on developing churn prediction and Life-Time Value (LTV) forecasting models specifically for the "Growth" and "Relationship" pillars. This would allow AIAI to intervene pro-actively before a high-value customer migrates to a competitor.
2. Temporal & Behavioral Integration: We recommend a secondary study into temporal travel patterns (e.g., seasonality and route preferences). Integrating these behavioral insights into the existing demographic framework would enable "hyper-personalized" marketing timing, such as offering family vacation packages to the Suburban Married segment exactly when their historical data suggests they begin planning their annual trips.

In conclusion, this segmentation framework provides AIAI with more than just a customer list; it provides a strategic roadmap. By focusing on scalability and differentiated service, AIAI is now positioned to transform its loyalty program from a cost center into a powerful engine for sustainable revenue growth.

8. ANNEX – FIGURES AND VISUAL OUTPUTS

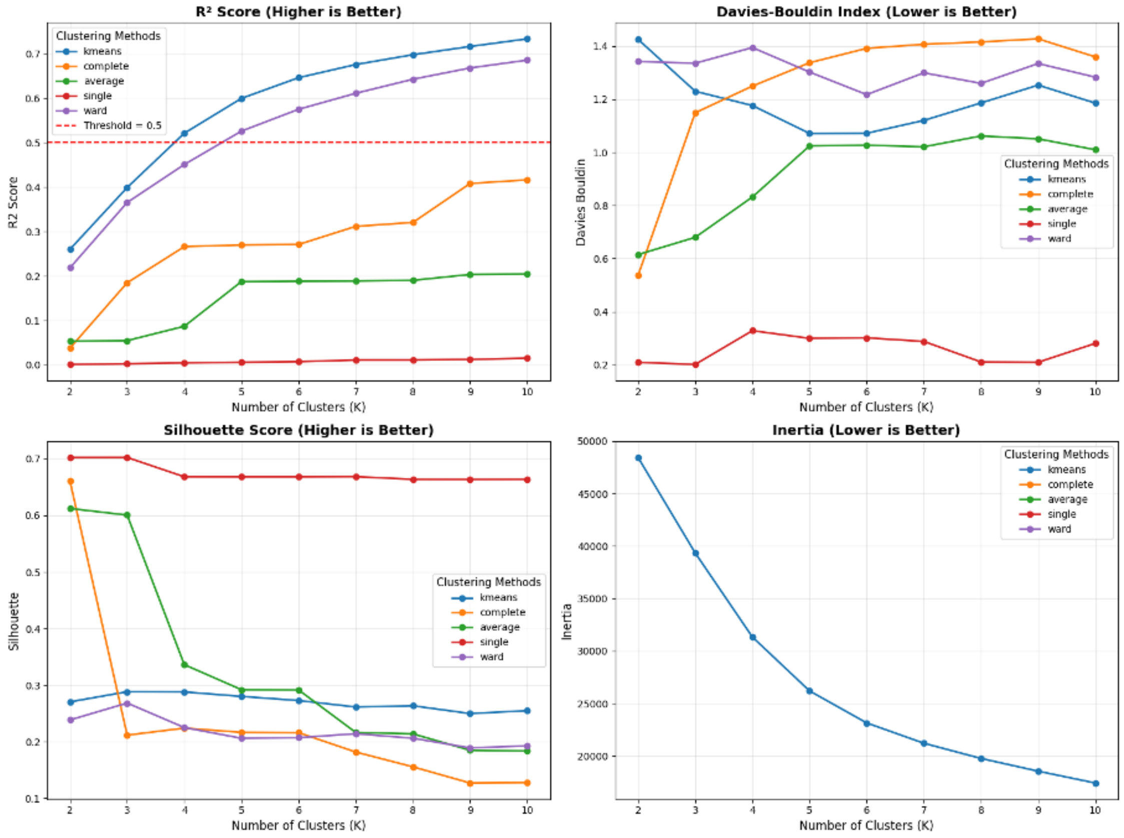

Figure 1 – Metrics – Value-Based Perspective

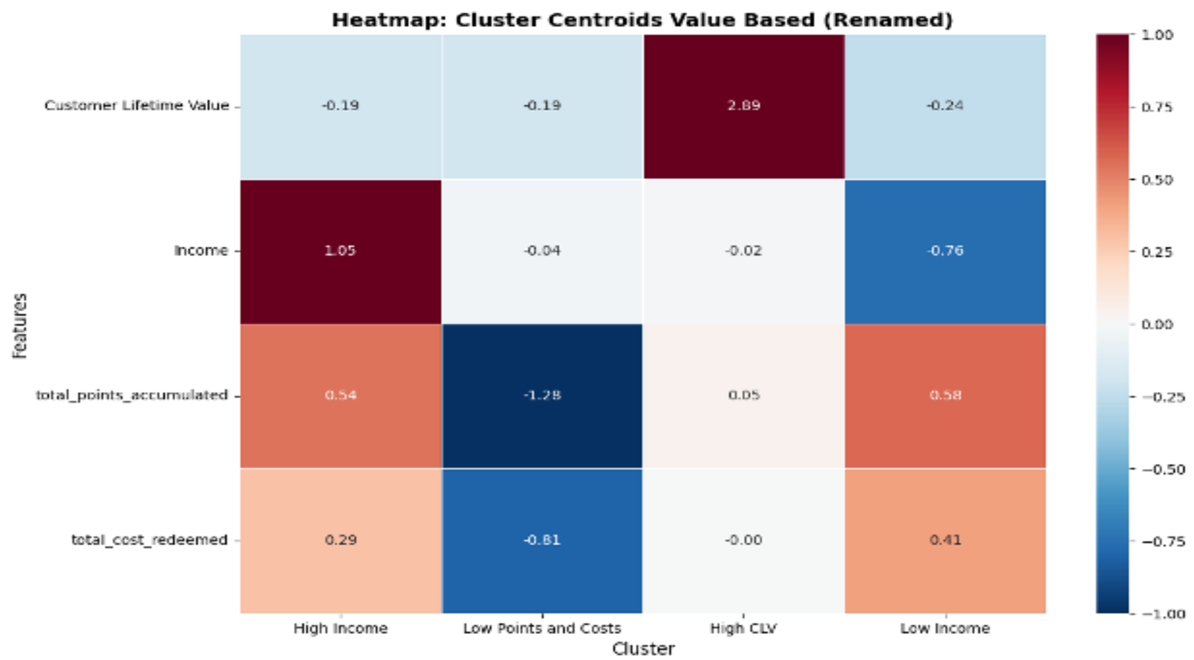

Figure 2 – Heatmap Cluster Centroids – Value-Based Perspective

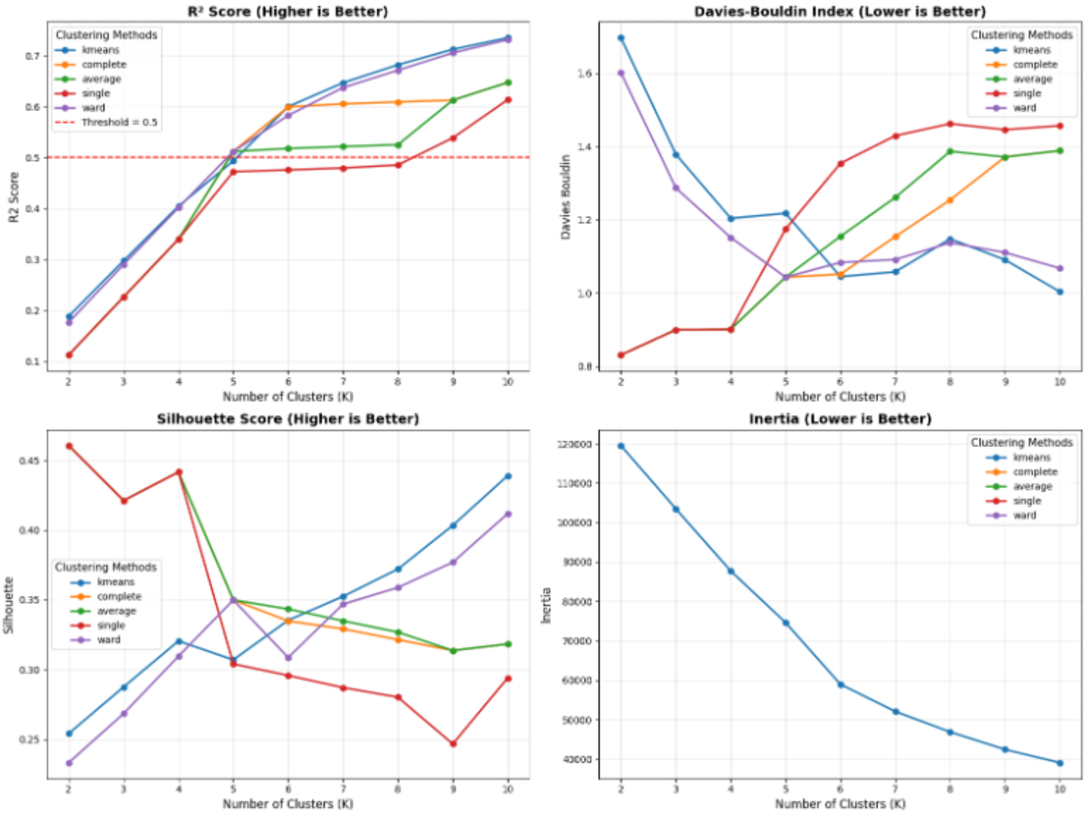

Figure 3 – Metrics – Demographic Perspective

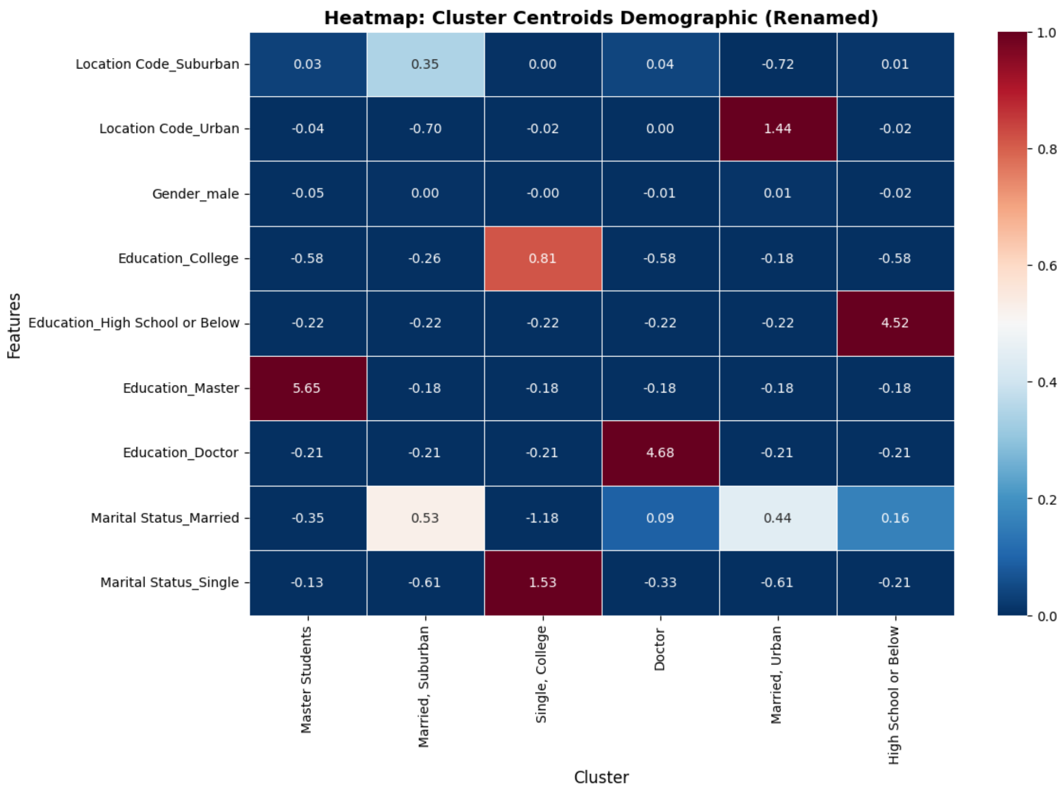

Figure 4– Heatmap Cluster Centroids – Demographic Perspective

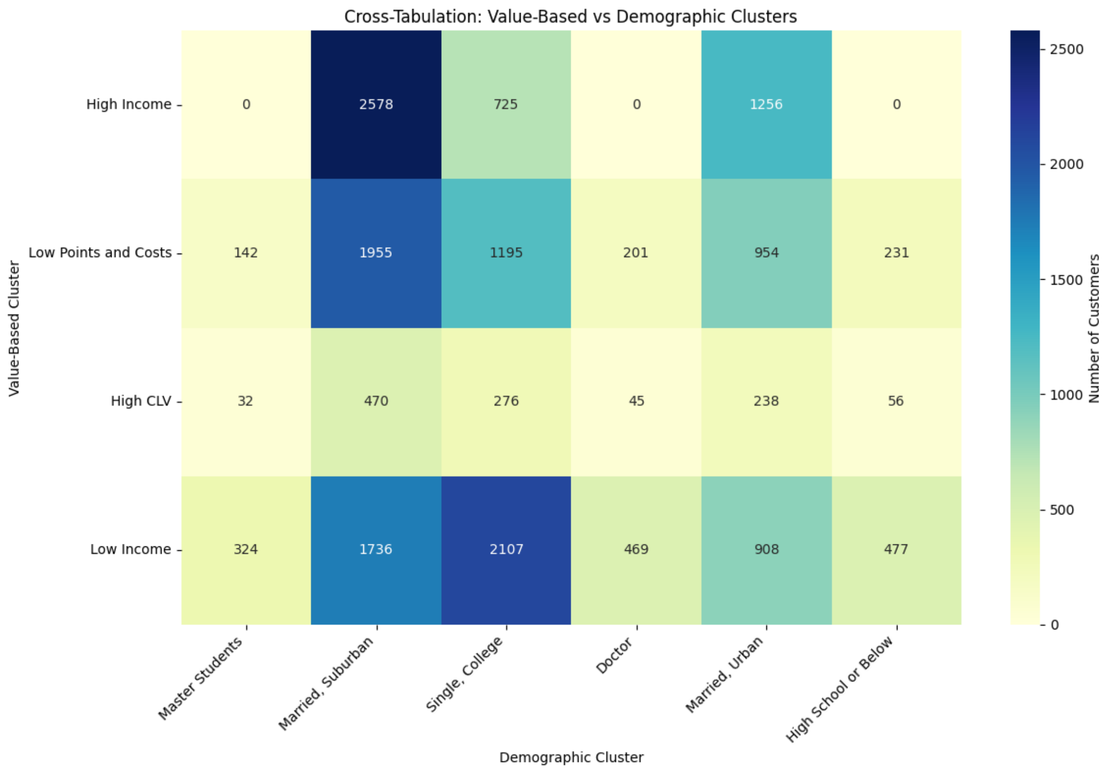

Figure 5 – Cross-Tabulation: Demographic vs Value-Based

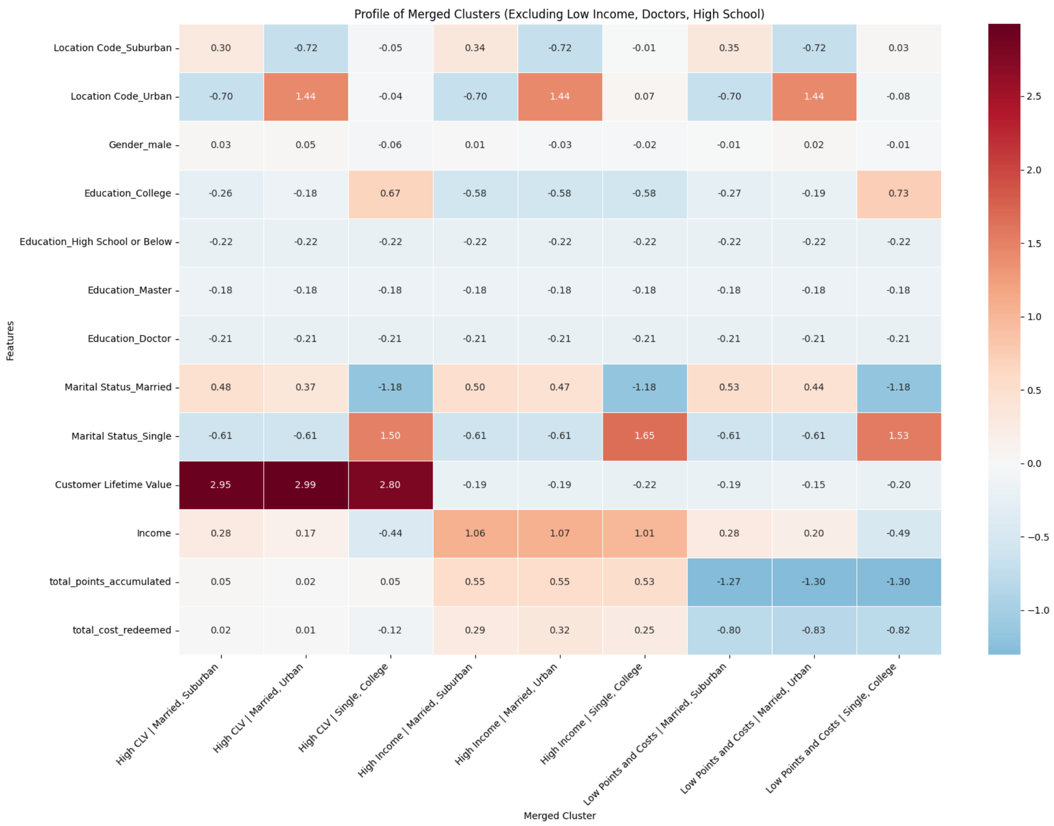

Figure 6 – Heatmap – Merged Cluster (Excluded features)

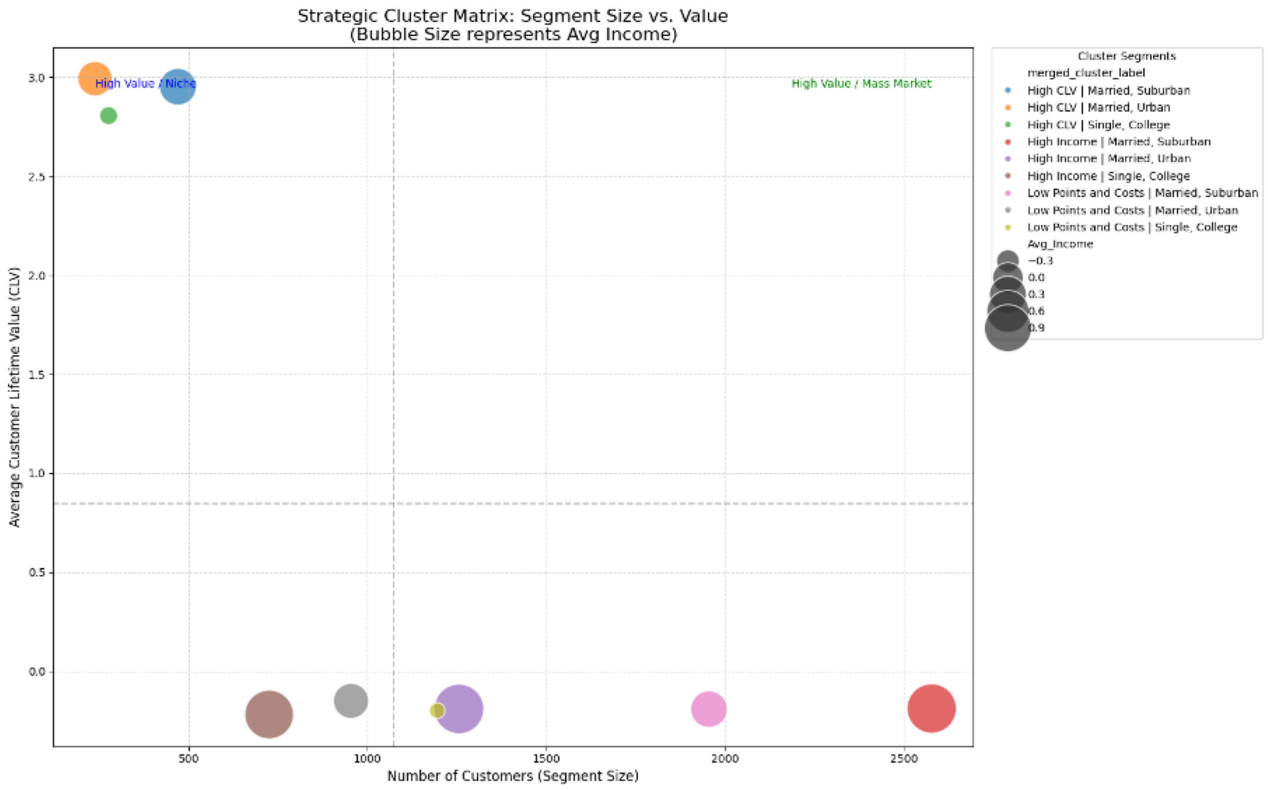

Figure 7– Strategic Cluster Matrix – Segment Size vs. Value

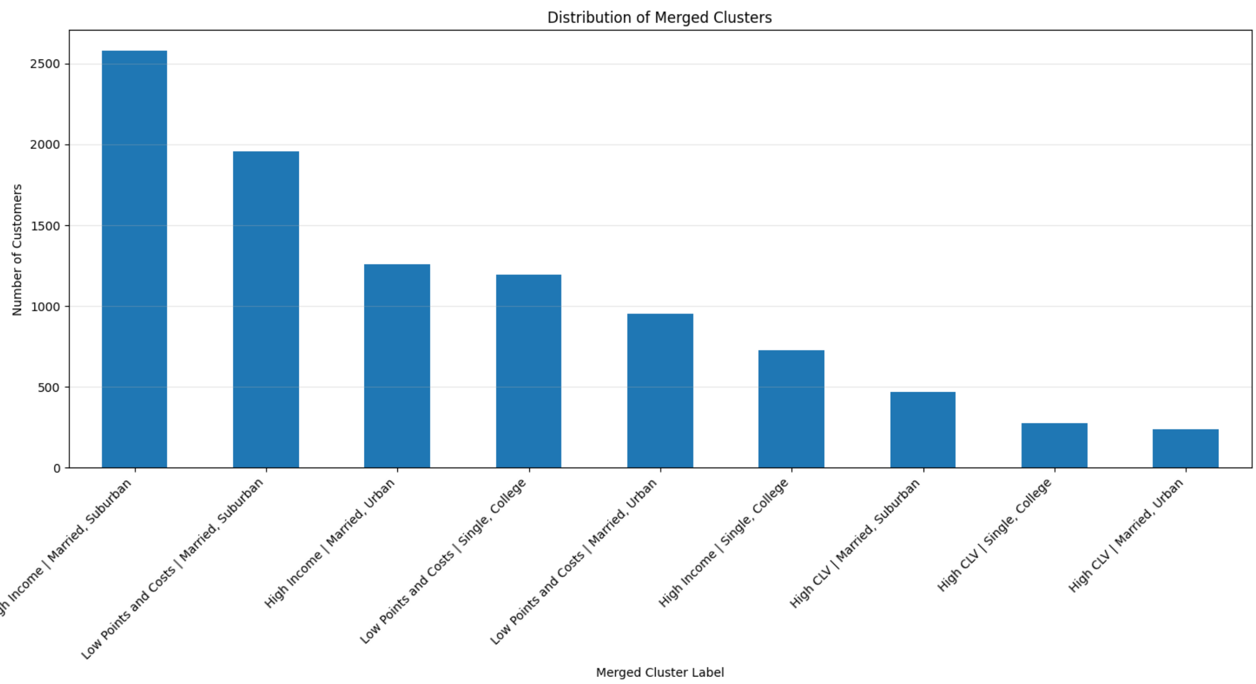

Figure 8 – Bar chart - Number of Clusters for Merged Cluster (Excluded Features)

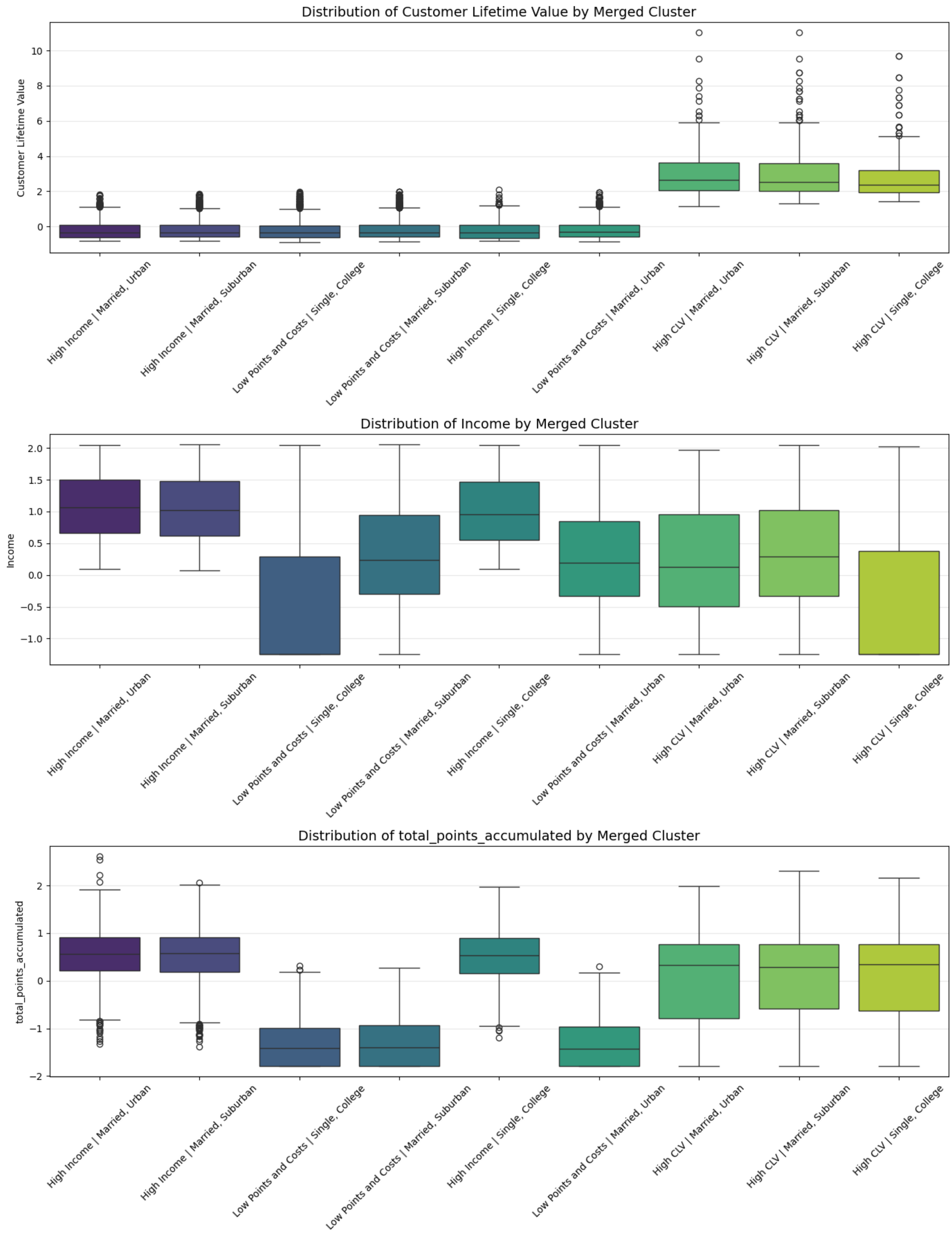

Figure 9 – Boxplots – Merged Cluster for CLV, Income and Total Points Accumulated

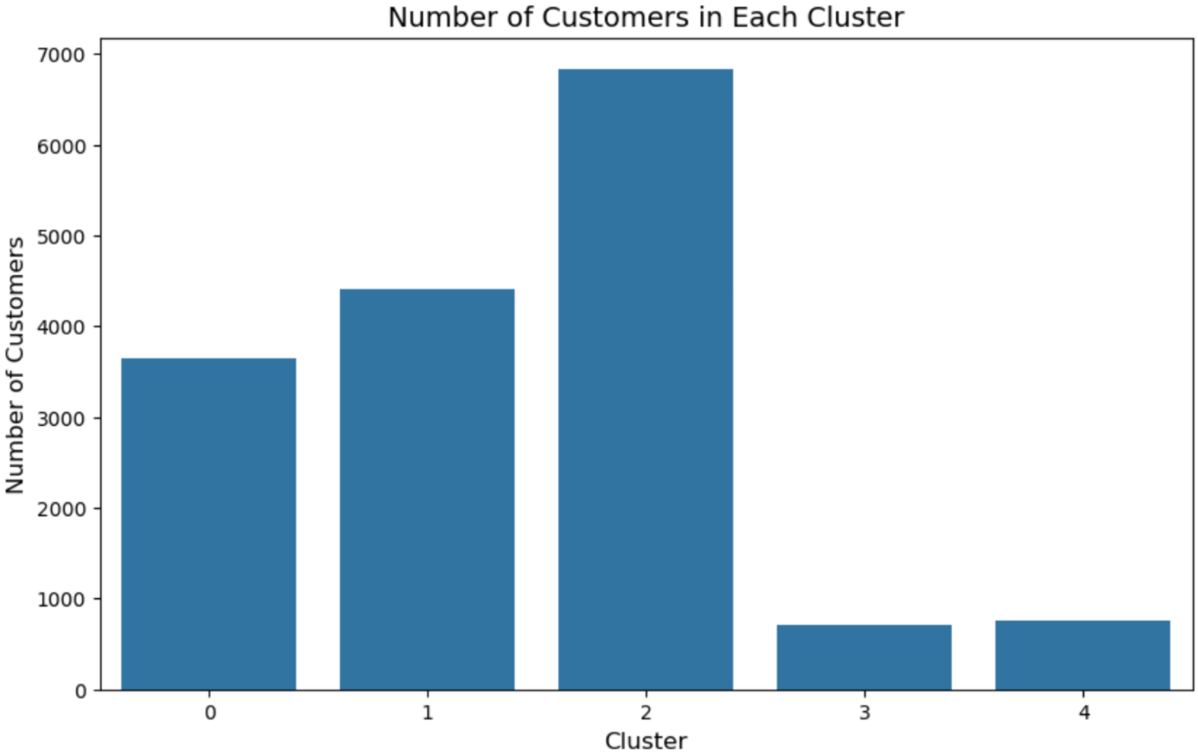

Figure 10 – BarChart – Number of Customers in each demographic Cluster

9. ANNEX – SUPPORTING STATEMENTS

9.1. Annex AI Usage Statement

Artificial Intelligence tools, specifically ChatGPT, were used to support code development and debugging, as well as to improve the structure and clarity of the written report. All coding logic, analytical decisions, model selection, result interpretation, and strategic conclusions were designed, reviewed, and validated by the project team. AI was used strictly as a supportive tool and did not perform autonomous analysis or decision-making.

9.2. Annex Contribution Statement

Individual student contribution to this deliverable.

Duarte Gomes (Student ID: 20250017)

- Demographic Clustering Analysis and Visualizations
- Strategic Recommendations
- Conclusion

Esra Salhi (Student ID: 20250537)

- Behavior Clustering Analysis and Visualizations
- Executive Summary
- Introduction

Mehmet Karaca (Student ID: 20250344)

1. Value-based Clustering Analysis and Visualization
2. Notebook Planning, Review and Organization
3. Methodology
4. Results and Validation

All members contributed to collaborative discussions and ideation.

9.3. Annex Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the results of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.

10. ANNEX DATASET INFORMATION

10.1. Dataset 1: DM_AIAI_CustomerDB.csv

Loyalty#: Unique customer identifier for loyalty program members.

First Name: Customer’s first name.

Last Name: Customer’s last name.

Customer Name: Customer’s full name (concatenated).

Country: Customer’s country of residence.

Province or State: Customer’s province or state of residence.

City: Customer’s city of residence.

Latitude: Geographic latitude coordinate of customer location.

Longitude: Geographic longitude coordinate of customer location.

Postal Code: Customer’s postal/ZIP code.

Gender: Customer’s gender.

Education: Customer’s highest education level (Bachelor, College, etc.).

Location Code: Urban, Suburban, or Rural classification of customer residence.

Income: Customer’s annual income.

Marital Status: Customer’s marital status (Married, Single, Divorced).

Loyalty Status: Current tier status in the loyalty program (Star > Nova > Aurora).

Enrollment Date Opening: Date when the customer joined the loyalty program.

Cancellation Date: Date when the customer left the loyalty program.

Customer Lifetime Value: Total calculated monetary value of the customer relationship.

Enrollment Type: Method of joining the loyalty program.

10.2. Dataset 2: DM_AIAI_FlightsDB.csv

Loyalty#: Unique customer identifier linking flight activity records to the customer database.

Year: Year of the flight activity record.

Month: Month of flight activity record (1–12).

YearMonthDate: First day of the month representing the activity period.

NumFlights: Total number of flights taken by the customer during the month.

NumFlightsWithCompanions: Number of flights where the customer traveled with companions.

DistanceKM: Total distance traveled in kilometers during the month.

PointsAccumulated: Loyalty points earned by the customer during the month.

PointsRedeemed: Loyalty points redeemed by the customer during the month.

DollarCostPointsRedeemed: Monetary value of points redeemed during the month.